Practical 7

In [9]:
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# ── Playfair Cipher ──────────────────────────────────────────────────────

# Step 1: Build 5x5 Key Matrix (I and J share a cell)
def build_matrix(keyword):
    keyword = keyword.lower().replace('j', 'i')
    seen, key = set(), []
    for c in keyword + string.ascii_lowercase:
        if c not in seen and c != 'j':
            seen.add(c)
            key.append(c)
    return [key[i*5:(i+1)*5] for i in range(5)]

def find_pos(matrix, char):
    for r, row in enumerate(matrix):
        if char in row:
            return r, row.index(char)
```

In [10]:
```
# Step 2: Prepare Text (split into digraphs)
def prepare_text(text):
    text = text.lower().replace('j', 'i').replace(' ', '')
    text = ''.join(c for c in text if c.isalpha())
    digraphs = []
    i = 0
    while i < len(text):
        a = text[i]
        b = text[i+1] if i+1 < len(text) else 'x'
        if a == b:
            digraphs.append((a, 'x'))
            i += 1
        else:
            digraphs.append((a, b))
            i += 2
    return digraphs
```

In [11]:
```
# Step 3: Encrypt / Decrypt
def playfair(text, keyword, mode='encrypt'):
    matrix = build_matrix(keyword)
    digraphs = prepare_text(text)
    result = ""
    d = 1 if mode == 'encrypt' else -1   # +1 encrypt, -1 decrypt

    for a, b in digraphs:
        r1, c1 = find_pos(matrix, a)
        r2, c2 = find_pos(matrix, b)

        if r1 == r2:                         # Same row → shift columns
            result += matrix[r1][(c1+d) % 5] + matrix[r2][(c2+d) % 5]
        elif c1 == c2:                       # Same col → shift rows
            result += matrix[(r1+d) % 5][c1] + matrix[(r2+d) % 5][c2]
```

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        else:                                   # Rectangle → swap corners
            result += matrix[r1][c2] + matrix[r2][c1]

    return result

# ── Demo ──────────────────────────────────────────────────────────────────────
import string

keyword = "monarchy"
msg     = "Hello World"

enc = playfair(msg, keyword, mode='encrypt')
dec = playfair(enc, keyword, mode='decrypt')

print("=== Playfair Cipher ===")
print(f"Keyword   : {keyword}")
print(f"Original  : {msg}")
print(f"Encrypted : {enc}")
print(f"Decrypted : {dec}")
```

```
=== Playfair Cipher ===
Keyword   : monarchy
Original  : Hello World
Encrypted : cfsupmvnmtbz
Decrypted : helxloworldx
```